

CAD, Simulation & 3D Printing

Design, test and prototype parts · Equigrow short notes

CAD workflow

- Sketch (fully constrained) → extrude / revolve → fillets & holes → assemblies → drawings.
- Design intent: dimension from meaningful references so changes update cleanly.

Simulation (FEA) basics

- Set material → constraints (fixed faces) → loads → mesh → solve → read stress and displacement.
- Factor of safety = material yield strength / maximum stress.
- Refine the mesh where stress is high; check results against hand calculations.

3D printing

- Slicer settings: layer height, infill, supports, orientation.
- PLA is easy to print; PETG is tougher. Leave about 0.2-0.4 mm clearance for parts that fit together.